Math 257 Definitions and Theorems

Week 14: Modules 44-46

Fall 2023

44 Principal Component Analysis

Setup:

- Given m objects, we measure the same n variables.
- Thus m samples of n-dimensional data $\rightsquigarrow$ $m \times n$ matrix (each row is a sample)
- Analyse this matrix to understand what drives the variance in the data.

Definition 44.1. Let $X = \begin{bmatrix}\mathbf{a}_1 & \dots & \mathbf{a}_m\end{bmatrix}^T$ be an $m \times n$ matrix. We define the **column average** $\mu(X)$ of X as

$$\mu(X) := \frac{1}{m}(\mathbf{a}_1 + \cdots + \mathbf{a}_m).$$

We say X is **centered** if $\mu(X) = \mathbf{0}$, and for X centered, we define the **covariance matrix** $\operatorname{cov}(X)$ of X as $\frac{1}{m-1}X^TX$.

Principal component analysis (PCA):

- Input: centered $m \times n$-matrix X.
- Compute $\operatorname{cov}(X)$.
- Since $\operatorname{cov}(X)$ is symmetric, we can find an orthonormal eigenbasis $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ of $\operatorname{cov}(X)$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$.
- Write $\operatorname{cov}(X)$ as a sum of rank 1 matrices:

$$\operatorname{cov}(X) = \lambda_1\mathbf{v}_1\mathbf{v}_1^T + \cdots + \lambda_n\mathbf{v}_n\mathbf{v}_n^T.$$

- Each principal component $\mathbf{v}_i$ explains part of the variance of the data. The larger λ_i, the more of the variances is explained by $\mathbf{v}_i$.

Remark. PCA using SVD.

- Let X be a centered data matrix. Observe that $X^TX = (m-1)\operatorname{cov}(X)$.

- To find orthonormal eigenbasis of $\text{cov}(X)$, it is enough to find orthonormal eigenbasis of X^TX.
- Compute the SVD of $X \rightsquigarrow X = U\Sigma V^T$.
- The columns of $V = \begin{bmatrix} \mathbf{v}_1 & \dots & \mathbf{v}_n \end{bmatrix}$ are the desired orthonormal eigenbasis.
- If σ_i is the singular value for $\mathbf{v_i}$, then

$$\lambda_i = \frac{\sigma_i^2}{m-1}$$

45 Review of Complex Numbers

Definition 45.1. $\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\}$ where $i = \sqrt{-1}$, or $i^2 = -1$.

- The **real part** of z, denoted $\Re(z)$ is defined by $\Re(z) = x$.
- The **imaginary part** of z, denoted $\Im(z)$ is defined by $\Im(z) = y$.
- The **complex conjugate** of z, denoted $\overline{z}$, is defined by $\overline{z} = x - iy$.
- The **absolute value**, or **magnitude** of z, denoted $|z|$ or $\|z\|$, is given by

$$|z| = \sqrt{x^2 + y^2}.$$

Remark. Any point in $\mathbb{R}^2$ can be viewed as a complex number: $\binom{x}{y} \leftrightarrow x + iy$.

Definition 45.2. Given $z = x+iy$, $w = u+iv$, we define $z+w = (x+u)+i(y+v)$, and

$$\begin{aligned} zw &= (x + iy)(u + iv) \\ &= xu + x(iv) + (iy)u + (iy)(iv) \\ &= (xu - yv) + i(xv + yu) \end{aligned}$$

Theorem 45.1. *Let $z \in \mathbb{C}$.*

- $\overline{\overline{z}} = z$
- $|z|^2 = z\overline{z}$
- $|z| = |\overline{z}|$

46 Complex Linear Algebra

Goal. Use complex numbers (instead of real numbers) as scalars.

Definition 46.1. The (complex) vector space $\mathbb{C}^n$ is of all *complex* column vectors $\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}$, where $z_1, z_2, \ldots, z_n$ are complex numbers.

Remark.

- Now multiplication by a *complex scalar* makes sense.
- We can define subspaces, Span, independence, basis, dimension for $\mathbb{C}^n$ in the usual way.
- We can multiply complex vectors by complex matrices. Column space and Null space still make sense.
- The only difference is the dot product, you need to use the complex conjugate to get a good notion of length:

$$\begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} \cdot \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} = z_1\bar{w}_1 + z_2\bar{w}_2 + \ldots z_n\bar{w}_n.$$

Definition 46.2. Let A be an $m \times n$-matrix. The **conjugate matrix** $\overline{A}$ of A is obtained from A by taking the complex conjugate of each entry of A.

Theorem 46.1. *Let A be a matrix with real entries and λ is an eigenvalue of A. Then $\bar{\lambda}$ is also a eigenvalue. Furthermore, if $\mathbf{v}$ is an eigenvector with eigenvalue λ, then $\bar{\mathbf{v}}$ is an eigenvector with eigenvalue $\bar{\lambda}$.*

Definition 46.3. Let A be an $m \times n$-matrix. The **conjugate transpose** A^H of A is defined as $\overline{A}^T$. We say the matrix A is **Hermitian** if $A = A^H$.